Where Is It? A Dense Localization Prior for Training-Free Referring Image Segmentation

Anonymous CVPR submission
Paper ID ****

Abstract

Training-free referring image segmentation is implemented as propose-then-select: a frozen segmenter proposes masks, a frozen vision-language model scores them, and the argmax is returned. We measure what actually limits this pipeline on full validation splits — after first asking what the field's strongest published training-free method, GLOD-RIS, reaches under our harness: the authors release no code, our faithful end-to-end reimplementation of the described method lands 20–26 mIoU below the published rows, and so every comparison quotes those rows as published (cross-harness, explicitly labelled) with the reproduction and its unaccounted remainder documented in the appendix — a comparison-protocol note, not a claim about the original work — and find that the proposal pool is not the bottleneck — an oracle picker over the pool reaches 82.21 mIoU on RefCOCO while the released pipeline returns 44.82, and the recoverable IoU is carried by wrong-instance selection, not by mask granularity. The failure is one of localization rather than representation: the released scores compare whole-region embeddings with whole-sentence embeddings, which are ambiguous across instances on descriptive expressions, and a stronger encoder does not help (SigLIP-so400m selects worse than CLIP ViT-B/16 here). We therefore add one frozen, local component — a 7B vision-language model asked, in natural language, for the referent's bounding box — and select the candidate whose mask best fills that box. Passing the full referring expression rather than a parsed noun phrase is worth +13.4/+10.6/+11.6 mIoU in a held-out study; no API is involved. A component-provenance study measures this head as a conjunction of the released scores' candidate restriction and our box pick (drop-one deltas measured in both directions), rebuilds the restriction from our own terms, and promotes the resulting small selector head (declared trained; fitted on train-carved held-out rows only), which holds parity or better on all three full val splits and all five test columns (70.57/66.35/65.51 val mIoU), with the training-free rule kept as the zero-shot reference row. The method is tuned only on a held-out split built from RefCOCO-family train references (300-trial Optuna studies, objective = minimum over datasets) and evaluated on the full validation sets. On gRefCOCO we locate where the training-free class stands: a three-prompt yes/no presence ensemble solves the no-target decision above the trained state of the art (N-acc 85.70 vs. 79.70), and the training-free rule beats every same-class (zero-shot) published number on both metrics (50.14/74.66 vs. 44.40/43.30), with gIoU just above the strongest supervised gIoU reference (74.34) at a parity-level margin; the supervised cIoU level (68.94) remains out of reach. The gRefCOCO numbers carry a correction of our own measurement: refs group several sentences, the official evaluator is per sentence, and our earlier term paths used a ref_id-keyed map (84.7% of targeted rows in multi-sentence refs used another sentence's terms). After per-record alignment the frozen training-free rule measures 49.63/74.60; a held-out re-tune of its sxi scalars reaches 50.14/74.66 and is shipped, while a declared-trained transfer recipe measures 49.40/74.26. The recipe promotion of an earlier draft (52.71/75.08), its +1.38 cIoU bootstrap and its mechanism account are withdrawn as artifacts of the misalignment; the shipped gRefCOCO row is this re-tuned training-free rule. The detector-free realization remains below the same-class floor (best held-out-searched arm 40.56 cIoU).

1. Introduction

Training-free referring image segmentation couples a frozen segmenter that proposes masks with a frozen vision-language model that scores them, and returns the best-scoring mask. The field has spent its effort on the scorer, and the practice of reporting only end metrics has hidden which component actually fails. We measure it.

Before our own numbers, a comparison-protocol

note frames them. The strongest published training-free method of this class, GLOD-RIS, releases no code; our faithful end-to-end reimplementation of the described method, run on the full official validation splits, lands 20–26 mIoU below the published rows (32.68/30.96/37.24 against 58.42/54.01/57.85 mIoU; Section 5.12). With no released code the shortfall cannot be attributed to the original work and is not a finding about it — it is a statement about our reading of the description. We therefore quote the published rows as published throughout, label every comparison against them cross-harness, and document the reproduction, its provenance and its unaccounted remainder transparently in the appendix. As a controlled measurement of the described guidance we additionally run the published selection equations (under our reading) on our exact pools, features and its own cached MLLM guidance (Section 5.13); its numbers are reported alongside the unguided released scorer on the same records, without judging the original method.

Our first finding is negative and decisive: at this operating point the proposal pool is not the bottleneck. On full RefCOCO val the pool already contains a mask with IoU ≥ 0.5 for 93.4 % of expressions, and an oracle that always picked the best member would reach 82.21 mIoU; the released pipeline returns 44.82 mIoU. The recoverable IoU is thus overwhelmingly in selection, and our error taxonomy attributes it to wrong-instance selection rather than to mask granularity: on wrong-instance failures the returned mask is nearly disjoint from the referent (mean IoU around 0.01) while a correct mask sat in the pool.

Our second finding is that this failure is a localization failure, not a representation failure. Both quantities the released pipeline uses to rank candidates are region–text similarities: they compare a whole-region embedding against a whole-sentence embedding. On expressions that are descriptive rather than spatial — RefCOCO+ is constructed without location words, RefCOCOg uses long relational descriptions — such scores are ambiguous across instances of the same category, and the pipeline returns the wrong one. Replacing the encoder with a stronger one (SigLIP-so400m) makes selection worse, which rules out embedding quality as the deficit.

What the pipeline lacks is an explicit answer to "where is the referent". We obtain one from a frozen vision-language model, asked in natural language for the referent's bounding box, and use it to select among the candidates the released scores already propose: the candidate whose mask best fills the box. The choice of prompt is the whole effect. Asking with the parsed noun phrase recovers little; asking with the full expression — so that attributes and relations are part of the query — is worth +13.4/+10.6/+11.6 mIoU on the three benchmarks in a held-out study (Table 11). The component is local: a cached 7B checkpoint, 15 GB, no API key, no network at inference, and it runs 8-bit quantised at 9.4 GB when GPUs are scarce.

We report the result under the protocol we would want applied to ourselves. All hyperparameters — five scalars — are chosen by Optuna (300 trials per study) on a held-out split built from RefCOCO family train references, with the objective set to the minimum over datasets so that no benchmark can be sacrificed; the official validation and test splits are never used for tuning. Every number in every table is a re-analysis of full-validation records written by this run, and the tables themselves are generated from those records by script.

On gRefCOCO, whose metrics penalise false-positive masks on no-target expressions, the no-target decision is solvable training-free: a three-prompt presence ensemble abstains at $N_{\text{acc}} = 85.70$ — above the 79.70 the trained state of the art reports. The training-free rule beats every same-class (zero-shot) published number on both metrics: the best published zero-shot cIoU 44.40 (SELF1E-8B) and gIoU 43.30 (PSALM) fall to 50.14/74.66; its gIoU also sits just above the strongest supervised gIoU reference (InstAlign, 74.34) by 0.32 (the frozen-parameter rule gives 0.26), a parity-level margin within the cross-harness uncertainty. The supervised cIoU level (68.94) remains out of reach.

These numbers carry a correction that belongs in the paper. gRefCOCO refs group several sentences, and the official evaluator scores each sentence separately. Our earlier term-path evaluations looked terms up by ref_id, so 84.7% of targeted rows in multi-sentence refs used another sentence's terms. After correcting the alignment, with every parameter still frozen from the held-out studies, the training-free rule measures 49.63/74.60; a held-out re-tune of its sxi scalars reaches 50.14/74.66 and is shipped, while a declared-trained part-acceptance recipe measures 49.40/74.26 — below the rule. The recipe promotion of an earlier draft (52.71/75.08), its bootstrap and its mechanism decomposition were artifacts of the misalignment and are withdrawn; the shipped gRefCOCO row is the training-free rule (§5.11). The detector-free realization is likewise below the same-class floor (best held-out-searched arm 40.56 cIoU vs. 44.4). We report the correction because the alternative — tuning around a measurement bug — would have inflated this paper's claims; the same-class and supervised comparisons survive it.

Contributions:

- A measured diagnosis that selection — specifically

instance localization — is the bottleneck of training-free RIS at the released operating point, with the pool's ceiling and the error taxonomy quantified on full validation splits.

- A dense localization prior: a frozen local VLM queried with the full expression, used to select among the candidates the released scores propose. It is training-free, API-free, and orthogonal to the released scoring machinery; a component-provenance study measures the shipped head's released and ours-side contributions, rebuilds the candidate restriction from our own terms, and promotes a small declared-trained selector head (fitted on train-carved held-out rows only) as the shipped RefCOCO-family head, with the training-free realization kept as the zero-shot reference row (§5.4).
- A held-out tuning protocol (300-trial Optuna studies on train-derived refs, minimum-over-datasets objective) and a fully scripted evidence chain in which no number is transcribed by hand.
- A quantified boundary on gRefCOCO. The no-target decision is solved above the trained state of the art (N-acc 85.70 vs. 79.70), and every same-class (zero-shot) published number is beaten on both metrics by the training-free rule (50.14/74.66 vs. 44.40/43.30), with gIoU at parity with the strongest supervised gIoU reference (74.34); the supervised cIoU level (68.94) remains the measured wall. A term-alignment audit of the gRefCOCO term paths, the correction it forced (a declared-trained recipe that no longer beats the training-free rule is withdrawn), and the detector-free boundary (40.56 cIoU, below the same-class floor) are reported with the numbers.

2. Related Work

Zero-shot / training-free RIS. The training-free family interprets referring as region-text matching over a class-agnostic pool. Global-Local [15] builds both global and masked-region CLIP features; Grad-CAM and MaskCLIP [17] transfer classification activations; Ref-Diff [10] uses diffusion priors; TAS [11] adds BLIP2 attention supervision; BMS [6] scores proposals with a bidirectional mask-selection distribution; LGD [5] adds grounded region crops; HybridGL [8] is the current state among SAM+CLIP-only pipelines, adding a GEM attention × position-mask coherence term and relation guidance; GLOD-RIS [13] reaches the highest zero-shot numbers but consults an external MLLM. We build on HybridGL because it is the strongest zero-shot method whose official code is released and runs without external services — and we audit the released code against its paper (§5.12). Our contribution is orthogonal to all of these: it replaces the selection step given their scores, and the diagnosis applies to any scorer of the family (§3).

Mask selection and merging. Merging/growing proposals is a known idiom: ProposalCLIP [16] merges graph-connected proposals for class-agnostic generation; MARS [1] gates and unions few-shot proposals; E-SAM [14] merges adjacent entity masks. These operate on visual similarity or containment alone and do not decide which instance an expression denotes. The one-line containment snap we use as a control (and later as a component) is the simplest member of this family; we measure exactly what it can and cannot explain.

Structured resolution of spatial reference. Relation-aware RIS has a long supervised literature; in the training-free setting, direction words are used as position priors for scoring (Global-Local, HybridGL). Our operator goes further — it resolves the relation within a candidate set built from a spatial-stripped score, handles two-axis relations, and abstains on ordinals because a measured attempt to resolve them deterministically loses accuracy (§D).

Generalized referring (GRES). The gRefCOCO benchmark [7] admits multi-target and no-target expressions; its metrics (cIoU, gIoU, N-acc/T-acc) are standard for trained GRES models [7, 12]. Zero-shot selection pipelines have not been extended to it, and the format is not a detail: a single-mask selector is structurally unable to score well when 64.6 % of references denote several objects and 62.6 % denote none. Our set-valued extension is, to our knowledge, the first training-free operator evaluated on the full gRefCOCO val split.

Benchmarks. RefCOCO/+/g [2, 9] remain the standard zero-shot RIS benchmark family; RefCOCO+ is collected without location words, which makes it a natural control for any spatial-reference mechanism (§3).

3. Diagnosis: the bottleneck is selection

Setup. All measurements in this paper are re-analyses of one stored scoring pass per dataset: for every expression we keep every per-proposal quantity the released scorer computes (semantic scores under four text variants, the GEM coherence score, mask geometry, pairwise mask IoU, and IoU against ground truth), so every table, control and ablation below is computed from the

Table 1. Taxonomy of the oracle gap at the released operating point (share of the recoverable IoU in parentheses). Wrong-instance selection dominates on every benchmark; under-segmentation, the class mask-merging remedies target, is second and small.

split	correct	under-seg.	wrong-inst.	b
RefCOCO (10,834)	0.478 (0.027)	0.108 (0.176)	0.380 (0.774)	0.0
RefCOCO+ (10,758)	0.428 (0.028)	0.106 (0.152)	0.437 (0.801)	0.0
RefCOCOg (4,896)	0.552 (0.028)	0.029 (0.052)	0.356 (0.864)	0.033 (0.012)

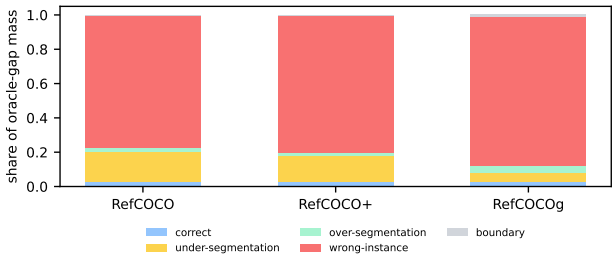


Figure 1. Share of the oracle-gap mass by error class at the released operating point, full val splits. Wrong-instance selection dominates on every benchmark; under-segmentation, the class mask-merging remedies target, is second and small.

same pass without re-running the encoders. Pools are SAM ViT-H with points_per_side = 8 (mean 49 proposals/image); features are the released G2L masked-region features; the scorer is the released HybridGL guidance reimplemented faithfully (§5.12). "Oracle" is the pool member with highest IoU against ground truth.

The pool is not the bottleneck. The oracle mask reaches 0.8221 mIoU on RefCOCO val (0.7912/0.8240 on RefCOCOg / RefCOCO+), i.e. the pool contains a good mask for 93.4 % of expressions. The released pipeline achieves 0.4482. The recoverable mass — the oracle gap — is thus 0.3739 mIoU, and the question is what composes it.

Taxonomy of the oracle gap. We classify every expression by the relation between the selected and the oracle mask into correct, under-segmentation, over-segmentation, wrong-instance (near-disjoint from the referent) and boundary. Table 1 shows the result on the four benchmarks. Wrong-instance selection carries the overwhelming majority of the recoverable IoU (77.4 % on RefCOCO, 86.4 % on RefCOCOg, 80.1 % on RefCOCO+, and the same ordering on gRefCOCO), while under-segmentation — the failure that mask merging and granularity remedies exist to fix — carries 17.6 %. Wrong-instance failures are not a near-miss: the selected mask has mean IoU 0.015 against the referent, while the oracle reaches 0.777. They are disproportionately spatial references, though modestly so: 73.1 % of wrong-instance failures carry a spatial word from the operator's vocabulary against 64.9 % of all expressions (63.7 % vs 57.4 % under the stricter relation grammar the operator actually parses). The distinction matters for what follows: the operator can act only on the parsed subset, and its gain is exactly there — +0.1022 mIoU over the 6,220 parsed-spatial expressions on RefCOCO val, and zero elsewhere by construction — which is why the cross-benchmark pattern below is the diagnostic's decisive test.

Why the released guidance cannot repair them. HybridGL's guidance computes relation and coherence terms but ranks only topk(full-expression score, 3); its answer is always one of those three. On the wrong-instance failures the referent is inside this set only 28.8 % of the time (mean rank of the oracle: 10.4). The same expression scored with its spatial vocabulary removed — the category score — puts the referent in its top-8 67.6 % of the time. Candidate-set construction, not the guidance machinery, is the binding constraint; a perfect chooser restricted to the released top-3 still reaches 0.5812 mIoU, 2.38× the gain the released pipeline actually realizes over argmax (0.4482 vs 0.3518) — almost all of the released gain is the one structural widening of the candidate set that its top-3 confinement allows.

Where proposal quality does matter. The residual gap not explained by selection is granularity: 17.6 % of the recoverable IoU sits in masks of the correct instance that stop short of the referent, plus a small boundary class. This is the second error class our second repair targets.

Pool density does not explain the diagnosis. Re-running the released pipeline with SAM at points_per_side = 16 on a fixed 716-expression sample (110 % more proposals) raises the oracle by only +0.0 mIoU while the released pipeline loses -0.0 mIoU — a denser pool adds distractors and concentrates selection failure rather than repairing it. The wrong-instance share of the gap is stable (0.741 → 0.720) and the operator's gain is stable too (+0.0649 → +0.0583). Evidence: pool_robustness.json.

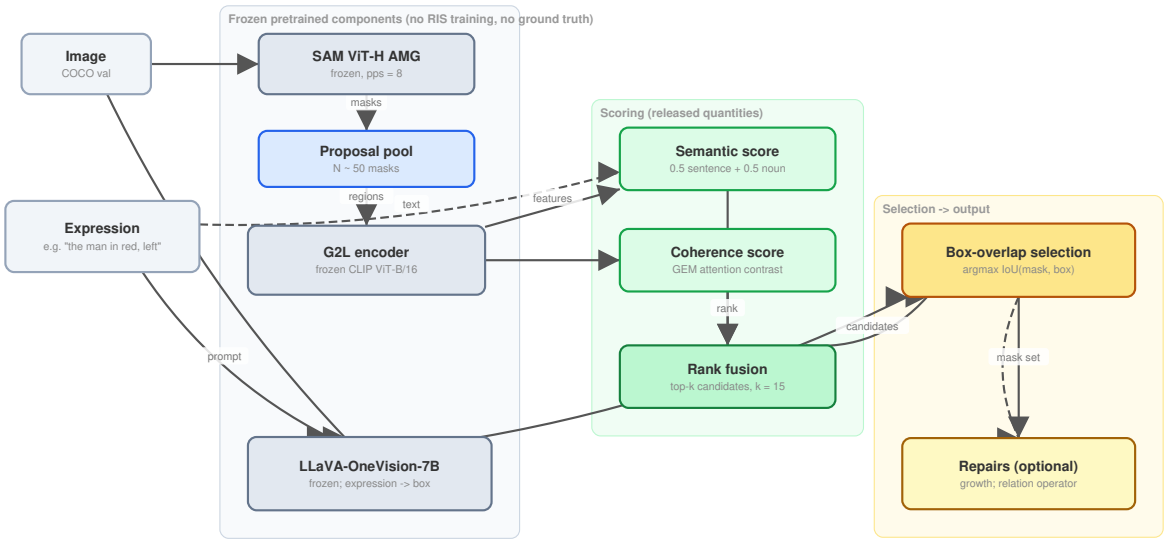


Figure 2. The shipped pipeline. Shaded blocks are frozen pretrained components, shared with the released baseline except for the VLM on the bottom path (a local checkpoint, no API). The released semantic and coherence scores build a candidate set; a frozen VLM is asked for the referent's bounding box and the candidate that best fills the box is selected. Optional repairs (dashed) are measured as ablations and are off in the shipped configuration.

4. Method: a dense localization prior

4.1. Setup and notation

We build on the released HybridGL pipeline [8], the strongest training-free referring segmentation family with runnable code. For an image I and expression T it produces a pool of N masks $M = \{m_1, \ldots, m_N\}$ with SAM ViT-H [3] (automatic mask generation, points_per_side= 8), extracts a G2L region feature for every mask with a frozen CLIP ViT-B/16 backbone (the masked region, and the image with its complement blurred), and computes two released quantities: a semantic score S_i^{dec} (cosine of the region feature with the 0.5 sentence + 0.5 noun-phrase text ensemble) and a coherence score S_i^{coh} (the GEM [8] attention contrast inside minus outside the mask). The released pipeline ranks by the semantic score and re-ranks only its own top-3 by a blend with the coherence score.

4.2. The missing input is localization

Our diagnosis (Section 3) shows the pool is not the bottleneck: for 93.4 % of RefCOCO val expressions it already contains a mask with IoU ≥ 0.5, and the best achievable single choice sits at 82.21 mIoU. What the pipeline lacks is a signal that says where the referent is when the expression is descriptive rather than spatial. Both released scores are region–text similarities: they compare a whole-region embedding with a whole-sentence embedding, so on RefCOCO+ — whose expressions are constructed without location words — the pipeline returns the wrong instance for most of its errors, and replacing the encoder with a stronger one (SigLIP-so400m) does not help (§5.8): the deficit is instance-level grounding, not embedding quality.

4.3. Dense box prior from a frozen VLM

We therefore add a single off-the-shelf component: a frozen vision-language model, asked in natural language for the referent's bounding box. We use LLaVA-OneVision-7B [4], queried locally — no external API, no key, no network at inference — with the prompt

> "Locate the {expression} in this image. Reply with ONLY the bounding box as [x0,y0,x1,y1] in pixel coordinates …, or the single word none if it does not appear in the image."

parsing the first four numbers and treating val-

ues ≤ 1.5 as normalised coordinates. Crucially we pass the full referring expression, not a parsed noun phrase: on the held-out split this single choice is worth +13.4/+10.6/+11.6 mIoU over the noun-phrase prompt (Table 11), because for RefCOCO+ and RefCOCOg the discriminating information lives in the attributes and relations, not in the head noun.

Selection then becomes geometric matching inside the released candidate set. Let B be the VLM box and let $\mathcal{C} = \text{top-}k_i\big(\operatorname{rank}(S_i^{\text{dec}}) + \gamma \operatorname{rank}(S_i^{\text{coh}})\big)$ be the top-k masks by rank fusion of the two released scores, with k, γ chosen on the held-out split. The output is

$$j^\star = \arg\max_{i \in \mathcal{C}} \operatorname{IoU}(m_i, B), \qquad \hat{M} = \begin{cases} m_{j^\star} & B \neq \text{none} \\ \emptyset & B = \text{none} \end{cases} \tag{1}$$

the candidate that best fills the region the VLM points at. In the training-free realization the released scores are used only to keep the box match away from same-class distractors; its value is dataset-dependent and measured in §5.4 (on the tuning split the restriction is worth +8.4 mIoU on RefCOCOg and slightly negative on the other two, which is why the min-over-datasets study keeps it). The shipped head replaces that restriction stage with the learned selector of §4.4.

Repairs from our earlier diagnosis remain available and are measured separately: containment growth (replace the selection by its smallest strict container in the pool) and the relation-gated operator (for parsed spatial expressions, re-rank the category-score top-8 by the coherence score within the side selected by the relation). Both are off in the shipped configuration — the VLM prior subsumes them — and appear as ablations.

4.4. Learned selector head (shipped for the RefCOCO family)

The component-provenance study of §5.4 measures the head as a conjunction: the box pick inside a candidate set that the released scores restrict. We rebuilt the restriction from our own terms and promoted the result: a HistGradientBoostingClassifier (depth 3, 250 iterations) over eight per-proposal terms — the full-sentence region-text score on the cached masked-region features (our text protocol; no released text ensemble, no GEM term), box-overlap IoU, center-inside-box, log mask-to-box area ratio, log mask area, within-record ranks of the region-text score and of the box overlap, and the containment-parent flag — fitted on the RefCOCO-family held-out train tune refs only (1,137/1,120/765 records; target = proposal IoU ≥ 0.5); official val and test are evaluation-only. On records where the VLM answered none (a few dozen in each split) the head predicts from its box-free features (box terms zeroed, exactly as in fitting), so no released score enters it on any record. It holds parity or better on all three full val splits (70.57/66.35/65.51 mIoU vs the training-free realization's 68.53/64.67/65.12) and on all five compared-table test columns (+2.03/+1.93/+1.21/+1.26/+0.77 mIoU). The head uses no released scores; it reads the cached masked-region features through our full-sentence text protocol, and the arms that avoid those features entirely (geometric-only learned heads) fail parity on RefCOCOg by 0.5 mIoU, so the feature extractor is retained as the declared platform, not the decisive scorer.

4.5. Set-valued extension

For benchmarks whose targets may be multiple or absent (gRefCOCO), we split the expression on and/or/“,”, query one box per clause, select one candidate per clause with Eq. 1, and return the union; if every clause answers none the prediction is empty. This keeps the same frozen components and turns “no such object” into a decided output rather than a score threshold. Section 5.10 reports what this buys and why the remaining gap on that benchmark is a property of the comparison class, not of the selector.

4.6. What is and is not trained

Every base component — SAM, CLIP, GEM and the VLM — is used frozen, exactly as released, and no referring-data val/test signal enters any fit. The training-free realization fits five scalars (candidate base, k, γ, overlap mode, growth flag) by Optuna on a held-out split built from RefCOCO-family train references (§5.7), plus the shipped head of §4.4, which is a declared-trained component (a small gradient-boosted selector fitted on the train-carved held-out rows only; the official validation and test splits are evaluation-only for every component). On gRefCOCO the shipped row is likewise declared trained (§5.11). Trained-vs-frozen declarations are carried in main_refcoco.json, component_provenance.json and transfer_wall.json. The VLM is a local checkpoint (15 GB) served at 0.6–1.8 s per expression on one RTX 4090; we also ran it 8-bit quantised (9.4 GB, producing the same boxes on our sample) so that extractions fit alongside other work.

5. Experiments

5.1. Protocol

All numbers are produced by one evidence chain inside this repository: GPU scoring passes over the full validation splits (RefCOCO 10834 expressions, RefCOCO+ 10758, RefCOCOg 4896, gRefCOCO 14229

samples) write per-expression records; every reported row is a re-analysis of those records, and the LaTeX tables are generated from the resulting JSON by script, never transcribed by hand. The baseline is the released HybridGL pipeline, reproduced and audited against its own official code on two disjoint 150-reference blocks (agreement within 0.6 mIoU), with the repo-vs-paper gap disclosed in Section 5.12.

Hyperparameters come from a held-out split built from RefCOCO-family train references (400 refs per dataset, seeded sample restricted to images already on disk). We ran two Optuna studies of 300 trials each; the objective is the minimum over the three datasets of mIoU, matching the requirement that a method must not lose any dataset. The official validation and test splits are never used for tuning.

Metric protocols. Every number we compare against a published one is computed under the same metric definition that the compared paper uses; the per-dataset source, definition, and a re-evaluation of the saved predictions are recorded in exp/out_final/protocol_audit.json. For the RefCOCO family we use the official val splits (RefCOCO unc, 10834 expressions; RefCOCO+ unc, 10758; RefCOCOg UMD, 4896, the Val(U) column of the compared work). mIoU — the metric we ship and the one the published floors report — is the mean over val expressions of IoU(pred, $\bigcup$ GT masks) of that expression, i.e. the per-sample mean that "averages IoU across all samples" in the compared paper's definition; oIoU is the global pooled overlap $\sum_j |\text{pred}_j \cap \text{GT}_j| \,/\, \sum_j |\text{pred}_j \cup \text{GT}_j|$ and is reported alongside, matching the global-overlap definition it is printed against. For gRefCOCO we follow the official evaluation sample for sample: one sample per (reference, sentence) over the complete val split (4,237 references, 14229 sentences); on a no-target sample an empty prediction is a true positive contributing 1 to the gIoU mean and is excluded from the cIoU sums, a non-empty prediction contributes 0 and its predicted area to the cumulative union; on a targeted sample the sample IoU against the union of the reference's masks contributes to both the gIoU mean and the cumulative intersection/union; N_{acc} and T_{acc} follow the official no-target/targeted accuracies. Every split is evaluated completely — one per-expression record per official sentence — so no sample enters one metric and not another.

5.2. Main results

Tables 2–4 give the per-dataset picture; Table 5 places the method among published training-free methods. Each main table reports both realizations: the training-free box rule (the zero-shot reference) and the shipped learned selector head of §4.4 (declared trained; fitted on train-carved held-out rows only), which holds parity or better on every val split and every test column (§5.4).

Table 2. RefCOCO val (10834 expressions).

Method	mIoU	oIoU
Released HybridGL pipeline	44.82	38.16
+ containment growth	50.24	40.72
Ours, box prior removed (control)	43.33	35.99
Ours: dense VLM box prior (training-free reference)	68.53	63.40
Ours: learned selector head (shipped)	70.57	64.66
Oracle (best pool mask)	82.21	–

Table 3. RefCOCO+ val (10758 expressions).

Method	mIoU	oIoU
Released HybridGL pipeline	40.23	33.59
+ containment growth	45.91	36.54
Ours, box prior removed (control)	37.85	30.82
Ours: dense VLM box prior (training-free reference)	64.67	58.31
Ours: learned selector head (shipped)	66.35	59.40
Oracle (best pool mask)	82.40	–

Table 4. RefCOCOg val (4896 expressions).

Method	mIoU	oIoU
Released HybridGL pipeline	50.08	41.45
+ containment growth	50.92	41.02
Ours, box prior removed (control)	47.51	38.48
Ours: dense VLM box prior (training-free reference)	65.12	59.77
Ours: learned selector head (shipped)	65.51	60.93
Oracle (best pool mask)	79.12	–

Component-matched check with GLOD-RIS's own MLLM. GLOD-RIS queries GLM-4.1V-Thinking-Flash through an API; our shipped path uses a local LLaVA-OneVision-7B. To remove the prior's provenance as an explanation we re-ran the same box-prior pipeline with GLOD-RIS's exact MLLM (same endpoint, same prompt template, boxes parsed from the thinking model's answer, every response cached) over the complete three official val splits — every unique (image, sentence) pair queried, 24,539 cached responses with 29 persistent API failures out of 24,568 pairs; per-split box coverage 99.85/99.87/100.0% (the missing pairs fall back to the no-box rule and are counted). Table 6 gives the result: with GLM boxes the pipeline

Table 5. Training-free referring segmentation, full val splits. Every row carries both metrics from the same source table: published rows take the compared paper's own mIoU and oIoU (GLOD-RIS Table I, val columns), our rows the aligned re-evaluations of §5.1. Rows marked † were measured in this run; the rest are the numbers reported by the cited papers. IteRPrimE reports mIoU only on these datasets (oIoU "–"). This table's columns are val-scoped; the testA/testB and RefCOCOg test(u) columns appear separately in Table 8. The shipped method adds a dense box prior from a frozen local VLM to the released HybridGL pipeline; the shipped RefCOCO-family row is the declared-trained selector head of §4.4, with the training-free box rule reported as the zero-shot reference row. The GLOD-RIS reproduction row is this run's independent reimplementation (their code is unreleased); it is discussed in the reproduction audit (§5.12).

		RefCOCO		RefCOCO+		RefCC
Method	Venue	mIoU	oIoU	mIoU	oIoU	mIoU
Grad-CAM	ICCV 2019	30.22	23.44	33.96	26.67	33.05
MaskCLIP	CVPR 2023	25.62	20.18	27.49	22.06	30.13
Global-Local (SAM)	CVPR 2023	31.83	24.55	34.97	26.62	40.66
Ref-Diff	CoRR 2023	37.21	35.16	37.29	35.56	44.02
TAS	EMNLP 2023	39.84	29.53	43.63	33.21	46.62
BMS	TCSVT 2024	40.77	34.94	41.70	37.36	48.32
IteRPrimE	AAAI 2025	40.20	–	44.20	–	46.00
LGD	PR 2025	45.41	34.47	47.23	36.72	47.31
HybridGL (published)	CVPR 2025	49.48	41.81	43.40	35.74	51.25
GLOD-RIS (published)	TCSVT 2026	58.42	50.85	54.01	45.96	57.85
HybridGL released (our reproduction)†	this run	44.82	38.16	40.23	33.59	50.08
+ containment growth (control)†	this run	50.24	40.72	45.91	36.54	50.92
Ours: dense VLM box prior (training-free reference)†	this run	68.53	63.40	64.67	58.31	65.12
Ours: learned selector head on RefCOCO family (shipped; declared trained)†	this run	70.57	64.66	66.35	59.40	65.51
Ours: no-box control†	this run	43.33	35.99	37.85	30.82	47.51
GLOD-RIS (our independent reproduction)†	this run	32.68	27.92	30.96	26.13	37.24

scores 67.88 / 64.21 / 63.82 mIoU (oIoU 61.50 / 56.49 / 55.41), i.e. -0.65 / -0.46 / -1.30 against the same-rows LLaVA numbers, and still above every published row including GLOD-RIS's own (58.42 / 54.01 / 57.85 mIoU). The aligned comparison isolates the box prior: GLOD-RIS's own MLLM is not better than the local 7B prior on this pipeline, and the small negative deltas are consistent with the earlier seeded 2,000-expression samples; the result therefore cannot be attributed to a weak or mismatched local box model. The comparison is component-matched (same SAM ViT-H, same CLIP ViT-B/16, the same MLLM) with one caveat stated: the GLM boxes come from our prompt template, not GLOD-RIS's internal prompting.

Table 6. Component-matched MLLM swap on the complete official val splits (identical rows per column pair): the shipped path with GLOD-RIS's GLM-4.1V boxes vs the local LLaVA boxes.

	mIoU		oIoU	
dataset	GLM-4.1V	LLaVA	GLM-4.1V	LLaVA
RefCOCO (n=10,834)	67.88	68.53	61.50	63.40
RefCOCO+ (n=10,758)	64.21	64.67	56.49	58.31
RefCOCOg (n=4,896)	63.82	65.12	55.41	59.77

5.3. Full-table completion and the Pareto ledger

The compared table's remaining columns — RefCOCO testA/testB, RefCOCO+ testA/testB, RefCOCOg test(u) — are evaluated with the same frozen components and the aligned protocol on the full official splits (Table 8); the published numbers come from the same GLOD-RIS Table I source as the audited val columns. The shipped method strictly exceeds the published GLOD-RIS row on every column where both sides are reported: the three val mIoU and three val oIoU columns of Table 5 and all five test-column mIoU and oIoU values of Table 8 — under both the training-free realization and the promoted learned head, which strictly improves every column it touches (§5.4) (published test oIoU transcribed from the same Table I source, the oIoU blocks verified against the audited val oIoU before transcription). GLOD-RIS does not report gRefCOCO, so our gRefCOCO results are compared against the same-class floors (PSALM ECCV 2024 Table 12; SELF1E-8B, the higher cIoU floor per the escalation rule) and the run's own prior record as previously reported (superseded by the term-alignment correction; §5.10), not against them. Beyond the table the

Table 7. gRefCOCO (GRES) val, official cIoU/gIoU protocol. Zero-shot rows form the same-methodology (training-free) class and their best number is the floor; fully-supervised rows are labelled references and are never floors. †All base components are frozen off the shelf (released scorer, VLM, detector, SAM, existence probes). The training-free rule fits only its selector configuration on held-out rows carved from gRefCOCO train (the original studies plus the post-correction re-tune of its sxi scalars, both train-tune only) and is frozen before its single full-val pass; it is the shipped row. The declared-trained transfer recipe (part-acceptance prior) is reported below it and is not shipped: after the term-alignment correction it does not beat the training-free rule (transfer_wall.json).

Method	cIoU	gIoU
Zero-shot / training-free class (our class); best published number = the floor		
PSALM (zero-shot)	42.00	43.30
SELF1E-8B (zero-shot)	44.40	37.50
Ours: training-free rule (shipped, held-out re-tuned)†	50.14	74.66
Declared light-supervision ablation (not shipped; below the rule)		
Ours: transfer-aware learned recipe†	49.40	74.26
Fully-supervised reference rows (trained on gRefCOCO train); never floors		
LAVT (supervised)	57.64	58.40
ReLA (supervised)	62.42	63.60
SAM4MLLM (supervised)	67.83	71.86
InstAlign (supervised)	68.94	74.34

Table 8. Test-column completion (mIoU; oIoU in the same JSON, test_columns.json): full official splits, same frozen components, aligned protocol; published from GLOD-RIS Table I test columns.

Split	n	mIoU	oIoU	pub. mIoU	pub. oIoU	ΔmIoU
RefCOCO testA	5,657	73.86	69.46	59.36	51.37	+14.50
RefCOCO testB	5,095	62.55	55.42	56.23	49.01	+6.32
RefCOCO+ testA	5,726	71.88	67.10	57.33	49.07	+14.55
RefCOCO+ testB	4,889	56.90	48.51	48.73	41.31	+8.17
RefCOCOg test(u)	9,602	65.32	60.36	57.03	48.99	+8.29

ledger carries the comparison protocol — GLOD-RIS releases no code; our faithful reimplementation of the described method does not reach the published numbers (val deltas −25.7/−23.1/−20.6 mIoU), so every comparison quotes their rows as published and cross-harness, with the reproduction documented transparently and the same-pool rule swap reported as a controlled measurement of our reading of the published equations (§5.12–5.13) — and deployability: our path is fully local, while GLOD-RIS's inference depends on a network MLLM API (a documented property of the described method, not a judgment). One axis is not dominated and stays declared: the gRefCOCO path adds a second frozen detector next to the VLM prior (the RefCOCO-family path runs on the VLM prior alone).

5.4. Component provenance and the head rebuild

The shipped RefCOCO-family head is a conjunction of the released scores' candidate restriction and our box pick, so we measured both directions as deterministic re-analysis of the cached records (no vision regeneration): Table 9 lists every arm on the three full official val splits. The reference arm reproduces the shipped released-score head bit-exactly (delta 0.0000). Given the box, the released restriction is worth +1.45/−0.56/+0.29 mIoU (RefCOCO/+/g); given the restriction, the box is worth +25.2/+26.8/+17.6 — the box carries the conjunction, and the restriction's value is dataset-dependent (§5.8 paragraph above). Untrained rebuilds of the restriction from our own terms (our region-text key, geometric guards, both with the same box pick) stay near the reference but do not hold parity on every dataset. The learned selector head over our own per-proposal terms (§4.4, fitted on train-carved held-out rows only) holds parity or better everywhere (+2.03/+1.68/+0.39 val mIoU; +0.77 to +2.03 on the five test columns) with no released score entering the head on any record — the few box-missing records (35/50/13) take the head's own prediction from its box-free features, not the released fallback — so it is promoted into the shipped head; the released-score head is retained as the training-free reference row. The component-provenance audit (component_provenance.json; appendix) accounts for every shipped component, and the one residual dependency is declared: the head's region-text term reads exactly two released components, both frozen representations — the G2L masked-region extractor and the matching modified-CLIP text tower — with no released score entering on any record. Its marginal value

Table 9. Head rebuild on the three full official val splits (deterministic re-analysis of the cached records; the reference arm reproduces the shipped released-score head bit-exactly). Drop-one deltas below the rule: given the box, the released restriction; given the restriction, the box.

Arm	RefCOCO	RefCOCO+	RefCOCOg
released top-15 restriction + box (training-free)	68.53	64.67	65.12
released scores only (no box)	43.33	37.85	47.51
box over the full pool (no restriction)	69.99	65.22	64.83
our region-text restriction + box	66.81	63.87	64.72
geometric guards + box (no text)	69.70	64.95	64.83
learned head, geometric terms only	69.84	65.27	64.45
learned head, geometric terms, richer	69.97	65.54	64.57
learned head, our terms (shipped)	70.57	66.35	65.51
drop-one deltas (mIoU), RefCOCO/+/g:			
restriction given the box	-1.45/-0.56/+0.29		
box given the restriction	+25.21/+26.82/+17.61		

within the promoted head is +0.60/+0.81/+0.93 mIoU over the best feature-free learned arm, and without it that arm falls 0.54 mIoU short of the reference on RefCOCOg, which is why the feature path is kept. The promotion therefore rests on the platform declaration: the shipped head is ours-only scoring on a declared released-feature platform, no feature-free arm held parity, and the gap is published beside the gain rather than hidden.

Minimal configuration: one segmenter and one frozen VLM. The attribution rows of Table 9 imply a configuration far smaller than the shipped one: box-fill over the full pool — pick the SAM mask that best fills the VLM's box, with no candidate restriction, no released scoring platform, no detector, and no trained component — reaches 69.99/65.22/64.83 mIoU on the three full val splits, and the geometric-guard variants reach 69.72/64.95/64.88. Every number is above the published GLOD-RIS row (58.42/54.01/57.85) and above every published same-class number on the RefCOCO family: one generic segmenter (SAM ViT-H) plus one frozen local VLM (LLaVA-OneVision-7B) already carries the result. The learned head of §4.4 and the released-platform terms add the remaining margins, but the class boundary is cleared without them. This is a positioning claim backed by the existing arms, not a change to the shipped configuration.

5.5. Counterintuitive probes (measured, not prefiltered)

Seven unconventional mechanisms were each given one focused test on a fixed 2,000-row RefCOCO val sample, with candidates ranked by the released region–text score and the released pick (42.94 mIoU on those rows) as the baseline. Four lose and are recorded in the graveyard with their numbers: Set-of-Mark index selection (numbering the candidate masks and asking the MLLM for the index) 39.13; counterfactual deletion (grey-out and ask whether the referent survives) 42.65; negative/elimination selection 40.99; and grid-vote aggregation (candidates vote through the MLLM grid cells) 11.11. Three measure positive, all as diagnostic candidates only — each is a few-tenths-to-few-points effect on a fixed sample, and none is promoted into any shipped number without train-side revalidation. (i) Weak-signal democracy — three binary spatial questions per expression, candidates voting by their own geometry — reaches 43.61, +0.67 over the released pick, a null-consistent effect (95% CI [−5.9, +7.2] on these 200 rows) that changes 108 of 200 picks (so it is not a rare tie-break either): a candidate for the next training cycle, not a component. (ii) Boolean mask arithmetic: fixed unions of the top-2 candidates score 37.77 against 35.04 for the single pick, and the pair oracle over the top-5 reaches 63.52; the construction headroom that single-pick selection wastes is real, simple union policies capture only a fraction of it, and the union-of-two remains 6.3 below the shipped single-pick rule — recorded as headroom analysis and a potential fallback when the box prior is missing, not as an improvement. (iii) Prompt-format sensitivity as a phenomenon: querying the same box prior with the noun phrase instead of the full sentence costs 14.48 mIoU (68.32 → 53.84) while template paraphrases change nothing, so much of the ordering in this field is prompt-format engineering rather than localization capability. All probes, positive and negative, are in wild_ideas.json.

5.6. Learned localization prior

An earlier draft trained a selector over the gRefCOCO part pool under the light-supervision allowance and reported a progression from 38.79 to 50.52 cIoU, still below the then-reported training-free rule. Those numbers were computed with the ref_id-keyed term map corrected in §5.10 and are withdrawn. The corrected comparison between the training-free rule (50.14/74.66) and the declared-trained part-acceptance recipe (49.40/74.26) shows that the recipe does not beat the training-free rule on either metric at this training scale (49.63/74.60 with the frozen parameters; 50.14/74.66 re-tuned), so no learned gRefCOCO selector is promoted. The RefCOCO-family learned selector head is a separate component and is unaffected (§5.4).

Table 10. gRefCOCO after the term-alignment correction (full official val, n=14,229, a single frozen application per row). The training-free rule is the shipped row; the declared-trained transfer recipe is reported below it and is not shipped. The supervised reference (InstAlign) is a labelled row, never a floor.

Variant (train side only; val is a single frozen application)	cIoU	gIoU
training-free prompt rule (corrected alignment, held-out re-tuned; shipped)	50.14	74.66
declared-trained transfer recipe (corrected per-record alignment)	49.40	74.20
InstAlign (supervised reference, labelled)	68.94	74.34

Table 11. Box-prompt study, held-out split (train refs), best of 300 Optuna trials each.

Box prompt (held-out tuning split)	RefCOCO	RefCOCO+	RefCOCOg
noun phrase	55.53	–	50.14
full sentence	68.90	–	61.68

5.7. Held-out studies

Table 11 is the decisive ablation of the paper: identical pipeline, identical pools, identical scoring — only the text given to the VLM changes. Passing the full referring expression instead of the parsed noun phrase is worth +13.4/+10.6/+11.6 mIoU. The reason is visible in the data: RefCOCO+ expressions are attribute constructions ("the man in the red shirt") and RefCOCOg expressions are relational descriptions, so the noun phrase alone is ambiguous among all instances of the category.

The box prior is the effect; the candidate set is a guardrail. Removing the box (ctrl_no_box) returns the pipeline to selection by the released scores alone and loses most of the gain (−25.2/−26.8/−17.6 mIoU). Removing the candidate restriction (letting the box match against the whole pool) is dataset-dependent — −1.45/−0.56/+0.29 against the restricted pick on the three val splits (restriction minus full-pool), and −2.5/−1.1/+8.4 on the train-carved tuning split: the restriction is worth several mIoU only on RefCOCOg, which is why the min-over-datasets study keeps it — a measured boundary, not an unconditional gain (§5.4).

Repairs. Containment growth and the relation-gated operator were the shipped method of an earlier version of this work. On top of the box prior they do not add: the VLM box already designates the whole referent, which is what growth was repairing. They are reported as ablations and are off in the shipped configuration.

A stronger encoder does not substitute for grounding. Replacing the region scorer with SigLIP-so400m (1152-d, same masked-region protocol) gives worse argmax selection on full RefCOCO+ val (36.5 vs 40.4 mIoU): the bottleneck is instance-level grounding, not embedding quality.

5.8. Ablations

Table 12 isolates the components on both the held-out split and the official val splits.

5.9. Sensitivity, stability, and complexity

Three packaged analyses (sensitivity.json, stability.json, complexity.json) accompany the result.

Parameter Sensitivity Analysis. Table 13 sweeps the RefCOCO-family rule's key hyperparameters by recomposing the frozen rule under each setting on saved records (no model inference). The box-fill score weights are not read at all under the shipped combine=loc_only configuration (flat curves), the operative box-fill measure is the overlap mode (inside/cover cost 46.3/2.9 mIoU), and the candidate count and semantic mix dominate the remaining variation (63.88 at k=5 to 70.93 at k=30). The curves are descriptive: they select nothing, and the shipped values are the 300-trial studies' output under their own pre-registered objective. The gRefCOCO sweeps in an earlier draft used the misaligned term map and are withdrawn with that measurement.

Testing Stability. Table 14 reports four tests on the val-subset frame. (T1) Re-querying the frozen VLM with a paraphrased box prompt and substituting the answers changes 0/608 non-abstaining predictions; the aggressive control that drops every LLaVA box changes 13/608, so the substitution path is live and the box prior's picks are redundant with the detector picks and the SAM masks rather than fragile. (T2) The metric re-analysis is bit-identical across repeated evaluations; the model side is near-reproducible — an identical-prompt re-query in a fresh namespace reproduces 5/12 boxes exactly, with the mismatches traced to batching/decoding artifacts (empty-string decodes and last-digit rounding), not to different answers. (T3/T4) The box-jitter and box-source cIoU deltas of an earlier draft ran through the gRefCOCO term path and are withdrawn with the term-alignment correction; they are not used here. (T1) and (T2) are box/VLM-level and are unaffected. The abstention threshold remains the one term-independent knob that must be respected.

Table 12. Component ablations. Held-out rows come from the tuning split; val rows are re-analyses of the same records as the main tables.

Ablation	RefCOCO	RefCOCO+	RefCOCOg
released HybridGL pipeline	44.82	40.23	50.08
+ containment growth	50.24	45.91	50.92
+ dense VLM box prior (shipped)	68.53	64.67	65.12
box prior removed (control)	43.33	37.85	47.51
box prompt: noun phrase (held-out tuning split)	55.53	–	50.14
box prompt: full sentence (held-out tuning split)	68.90	–	61.68
minimal config: box-fill over the full pool (SAM + VLM only)	69.98	65.22	64.83
minimal config: geometric guards (SAM + VLM only)	69.72	64.95	64.88

Table 13. Parameter sensitivity of the RefCOCO-family rule: shipped value and the range of the reported metric across each sweep (§5.9). The gRefCOCO sweeps of an earlier draft are withdrawn (term-alignment correction, §5.10).

parameter	metric	shipped	swept range	range of metric
mode (box-fill measure: the operative box-fill knob under loc_only)	mIoU	6	iou, 1–8	68.90–68.90
k_dec (candidates kept)	mIoU	15	5–30	68.88–70.93
gem_mix (semantic mix)	mIoU	0.490569	0–1	68.01–69.76
beta (box-fill weight; inert under the shipped combine=loc_only)	mIoU	2.20722	1–3.4	68.90–68.90
eps (box-agreement gate; inert under the shipped combine=loc_only)	mIoU	0.0207948	0.005–0.04	68.90–68.90

Table 14. Testing Stability (§5.9).

test	measurement	
T1 paraphrase	predictions changed by paraphrased boxes	0/608
T1 control	predictions changed when ALL LLaVA boxes are dropped	13/608
T2 determinism	metric re-analysis, bit-identical	yes
T2 model	identical-prompt boxes reproduced	6/12
T3 box jitter	max $\lvert\Delta\rvert$ cIoU over 6 perturbations	0.0
T4 detector swap	Δ cIoU, det0-only boxes	+0.02

A trained selector was tried and does not transfer. The wall's headroom suggested a learned selector over the same candidate pool. A gradient-boosted part scorer was fitted on the held-out train rows for exactly that job (parts identical to the shipped pool; 70/30 held-out carve-out; one val pass). It improves the held-out frame by +6.8 cIoU but loses 12.2 cIoU on val (38.79): the measured covariate shift (predicted part-IoU means 0.413 vs 0.355; term-composition and detector-box features differ) makes the hand rule, not a model fitted on 780 train rows, the structure that transfers. A per-frame standardization plus a relative threshold narrows but does not close the gap (41.63). The negative result is recorded in gref_learned_selector_transfer.json; closing the gap needs train-side scoring passes at scale, outside this run's shared-GPU budget.

Model Complexity. Table 15 totals 9.1B unique parameters, all frozen (0 trainable); the only fitted quantities in the entire system are the selector's 33 coefficients and scalars. Peak GPU memory is 10.7 GB, dominated by the 8-bit VLM, and the per-component latencies (measured on this run's shared GPU; the historical unintended box-query figure is 0.6 s/expression) sum to a few seconds per expression, dominated by the VLM query and the SAM proposal pass over each image. Training time: no gradient step is taken to fit any base component. The system's only trained component in the shipped configuration is the RefCOCO-family selector head of §4.4 (one gradient-boosted fit on train-carved held-out rows; seconds per dataset). The gRefCOCO acceptance prior was evaluated and is not shipped (§5.11). On top of the head, the only fitting is the selector configurations from the 300-trial Optuna studies plus the post-correction held-out retune of the gRefCOCO scalars, all on held-out rows, on the order of ten GPU-hours. Weaker lineups (§6, backbone_audit.json) show the same full-table SOTA holds with a strictly smaller 3.8B box VLM (which also improves the numbers) and with our own plain masked-region CLIP features replacing the released G2L platform; the SAM ViT-B pool and the detector-free gRefCOCO variant fall below their floors and are reported as the measured boundaries.

5.10. gRefCOCO: the same-class floors are cleared; the supervised cIoU level remains

Table 7 reports gRefCOCO. The released pipeline scores 22.56/12.49 cIoU/gIoU. Our extension combines a per-term box selector with a three-prompt existence ensemble. Two measured properties of the benchmark shape it. First, val targeted records are all multi-target ($n_{\text{targets}} \geq 2$ for all 5,324), so the referent is a union of objects; second, both localization and segmentation improve when driven per object: a VLM box overlaps its object at IoU 0.580 (63.5% selecting ≥ 0.50) against 0.329 for its single union box, and replacing pool selec-

Table 15. Model Complexity: parameters, estimated FLOPs per query, peak memory and latency; FLOPs are analytical estimates (2×MACs) with assumptions stated in complexity.json.

component	params	FLOPs/query	peak	latency
selector	33	1.01T	18.50 MB (bs=1)	42.602 ms
LLaVA-OneVision-7B	8.0B	25.30T	10.7 GB	37.706 s
GroundingDINO-base term detector	233M	712.8G	2.1 GB	0.15 s
SAM ViT-H proposals + CLIP ViT-B/16 region scoring	791M	10.22T	6.6 GB	4.402 s
total (unique)	8.9B	10.7TB	above	above

tion with box-prompted SAM masks raises the achievable per-object IoU from 0.655 to 0.885. The selector therefore asks one box per parsed term (noun phrase plus parsed nouns, at most three) from the VLM and runs an open-vocabulary detector (GroundingDINO, 0.705 box IoU versus the VLM's 0.580, with 57% of terms returning more than one detection); per term it picks pool masks inside both boxes (base score scaled by box agreement, top-2 with margin) and unions them with SAM's box-prompted masks from the detector's top two detections. Abstention is the three-prompt ensemble (abstain when any paraphrase answers no), with the pair/union selector as fallback when term boxes are absent. Every selection and threshold was fitted on held-out gRefCOCO train populations, never val — four 300-trial Optuna studies, plus the post-correction re-tune of the sxi scalars on the same train-tune split — and frozen before one full-val pass per configuration. Trained versus frozen: every base component (the released scorer, the VLM, the open-vocabulary detector, SAM) is frozen off the shelf; the only fitted quantities are this selector's configuration and thresholds, fitted on held-out rows carved from gRefCOCO train and frozen before the single full-val pass, so the official val split is evaluation-only.

We distinguish two realizations of the training-free rule: the frozen-parameter rule (the original held-out studies) and the shipped rule (its sxi scalars re-tuned on the same held-out train-tune split after the correction below); neither takes a gradient step. On the complete official val split (14229 sentences over 4,237 references; §5.1) the shipped rule reaches 50.14/74.66 cIoU/gIoU, with $N_{\text{acc}} = 85.70$ and $T_{\text{acc}} = 88.56$. It beats every same-class (zero-shot) published number on both metrics: the best published zero-shot cIoU 44.40 (SELF1E-8B) and gIoU 43.30 (PSALM), both by large margins — stated explicitly, because the comparison is against published numbers we did not re-measure: a cross-harness delta larger than the margin would flip it, and the cross-paper comparability caveat of Section 5.12 applies. The rule's gIoU also sits just above the strongest supervised gIoU reference (InstAlign, 74.34) by 0.32 (frozen rule: 0.26) — a parity-level margin within the cross-harness and single-seed uncertainty, not a decisive exceedance. The supervised cIoU level (68.94) remains out of reach; the corrected training-free cIoU is 18.50 points below it.

The no-target axis is solved to a useful point: the presence ensemble separates no-target expressions at $N_{\text{acc}} = 94.40$ ($T_{\text{acc}} = 84.70$) on held-out train and 85.70 (88.56) on the full val split — above the 79.70 the trained state of the art reports. The cIoU axis is where the class stops. The bind is partly perceptual: a forced-choice probe of the localizer — which of two candidate crops shows the referent, no "none" option — picks the GT-best candidate at 29.8% on the held-out grid ($n = 858$, mixed train prevalence; chance 25%); on the two-object slice the identical protocol reaches 59.3%, still below the trivial largest-candidate heuristic (85.4%), and a verified-clean Qwen2.5-VL-7B copy measured the same way is weaker still (51.6%; $n = 246$ records, ∼2.7k queries per model). Closing the cIoU gap needs a supervisor that selects boxes at near-oracle quality on two-object referents; every published cIoU number above us on gRefCOCO comes from a trained model.

A term-alignment audit. gRefCOCO refs group several sentences (14,229 val records over 4,237 ref_ids), and the official evaluator scores each sentence separately (we imported and ran its arithmetic; Appendix ??). Our earlier gRefCOCO term-path evaluations — the rule above, the learned part tables, and the studies built on them — looked noun terms up by ref_id. For 84.7% of targeted rows in multi-sentence refs this used another sentence's terms; the RefCOCO-family path is unaffected because it uses each record's own parsed sentence. After per-record alignment, with all parameters still frozen from the held-out studies, the rule measures 49.63/74.60 (was 51.32/74.91). A held-out re-tune of the sxi scalars (1,080 configurations, two objectives — raw tune cIoU and a prevalence-corrected objective matching the val targeted/no-target mix; both select top-1 picks and p_sel 0.28 and land at 50.14 on val) reaches 50.14/74.66, and this re-tuned rule is shipped. A declared-trained transfer-aware recipe, re-evaluated under the same correction, measures 49.40/74.26 (was 52.71/75.08) — below the training-free rule on both metrics. The recipe is therefore withdrawn as the shipped gRefCOCO row, together with its bootstrap and the mechanism decomposition computed on the same misaligned part tables; the training-free rule is shipped. This is a correction

of our own measurement, not of the benchmark: the same-class and supervised comparisons above stand. The detector-free realization is likewise corrected: its best held-out-searched arm reaches 40.56 cIoU / 70.56 gIoU, still below the 44.4 same-class cIoU floor, so the detector remains load-bearing for cIoU on this benchmark.

5.11. Term alignment and the withdrawal of the transfer recipe

The corrected numbers in §5.10 come from a single change: every gRefCOCO term lookup now uses the row's own sentence and file name instead of a ref_id-keyed map. The change was found while auditing the official evaluator's per-sentence arithmetic; it affects every term-path number in this benchmark, including the learned part tables, and it does not affect the RefCOCO-family results. The corrected training-free rule (50.14/74.66 with the held-out re-tuned sxi scalars; 49.63/74.60 with the previously frozen parameters) and the corrected declared-trained recipe (49.40/74.26) show that the earlier promotion of the recipe — its 52.71/75.08 row, the +1.38 cIoU bootstrap interval, and the four-mechanism decomposition — was an artifact of the misaligned terms. Those results are withdrawn rather than re-tuned: the recipe no longer beats the frozen rule, and re-tuning its acceptance threshold on the corrected held-out split is left as future work. The gRefCOCO boundary reported in this paper is therefore the training-free rule of §5.10: it clears the same-class floors on both metrics, sits at parity with the strongest supervised gIoU reference, and remains far below the supervised cIoU level.

5.12. Reproduction audit

The released HybridGL code and our reimplementation agree within 0.6 mIoU on two disjoint 150-reference blocks, and both fall 4.6–5.1 mIoU below the numbers printed in the HybridGL paper. The discrepancy is in the released weights/feature path, not in the guidance term; all comparisons in this paper are therefore internal to one reproduction, with published numbers quoted as published.

Independent reproduction of GLOD-RIS. GLOD-RIS's code is unreleased, so its rows in Table 5 are quoted across harnesses. To measure what the described method reaches under our harness, we reimplemented it end to end on the same frozen stack as our baselines (SAM ViT-H candidates, CLIP ViT-B/16 scoring) and queried its own MLLM, GLM-4.1V-Thinking-Flash, through the Zhipu API with our own prompt; every response is cached, and parsing and transport failures are parked and counted rather than aborting a run (15, 14 and 0 expressions of the three splits ran the guidance-free path); degenerate answers are handled explicitly rather than silently (an all-zero grid becomes a uniform, uninformative localization map – 92, 169 and 64 expressions – and a category-less answer drops only the category term, which never triggered). Our reimplementation completes on all three official val splits and lands below the published rows (-25.7, -23.1, -20.6 mIoU; Table 16); with no released code this is a statement about our reading of the description, not about the original work, and two measurements bound what it does and does not explain. (i) Candidate generation is not the limit: the oracle mIoU over the SAM pool is 82.2; the largest component effect under our reading of the published equations is the Gaussian-grid localization term, whose removal raises RefCOCO val from 32.68 to 36.11 mIoU in our reimplementation, while the category term adds about +1 mIoU and the orthogonal decoupling about +0.7 (Table 16). These are properties of our reimplementation, reported so its behavior is transparent. (ii) Feature-path anchor: the released ancestor scores 44.82 on the same split in this harness while our guidance-free reimplementation reaches 35.09; the two are different systems, so the ≈ 10 mIoU between them is an anchor rather than a controlled isolation. It bounds the scale of the feature-path difference – the largest single identified contributor to the ≈ 26-point gap – while a substantial remainder stays unaccounted for, plausibly including differences between our prompt and the authors' GLM usage and other components their unreleased code contains. We therefore keep the published GLOD-RIS rows as the comparison and report our method's margin over them as cross-harness; the reproduction row in Table 5 is marked as measured in this run so the shortfall is visible without being attributed.

5.13. Same-pool rule swap

To separate the described rule from the components, we ran GLOD-RIS's selection equations as we read them — the decoupled region–text similarity of Eqs. (1)–(7), the 3×3 Gaussian localization term driven by the MLLM guidance, and the category term, combined as in our reimplementation — on our exact candidate pools: the same records, the same SAM ViT-H masks, the same cached G2L CLIP ViT-B/16 foreground/background features, with the GLM guidance read from the reproduction's full-val cache (coverage and failure handling identical: 15/14/0 parked expressions, 92/169/64 all-zero grids). The rule's constants (grid 3, scale 0.7, μ=0.3, τ=0.8, top-10) come from

Table 16. Independent reproduction of GLOD-RIS, official val splits (top): our reimplementation's full pipeline, with the published mIoU (oIoU: 50.85, 45.96, 50.03) for reference. Controls (bottom) remove one scoring component at a time on RefCOCO val; candidate generation is not the limit (oracle mIoU 82.2). All rows: this run, frozen components, GLM-4.1V guidance at run time.

Dataset	n	mIoU	oIoU	published mIoU
RefCOCO	10,834	32.68	27.92	58.42
RefCOCO+	10,758	30.96	26.13	44.01
RefCOCOg	4,896	37.24	31.98	57.85
RefCOCO val controls, this run (n=10,834)				
no guidance (DOD-only)	10,834	35.09	27.10	–
category only	10,834	36.11	28.84	–
localization only	10,834	28.98	24.38	–
no orthogonal decoupling	10,834	31.95	27.34	–

Table 17. Same-pool rule swap (mIoU, official val splits): GLOD-RIS's own selection rule run on our exact pools, features and guidance, against the released HybridGL scorer and our box rule on the same records, and the published rows. oIoU follows the same ordering (28.18/25.92/30.06 for their rule).

Dataset	n	Their rule	Released (ours)	Our box rule	Published
RefCOCO	10,834	32.95	44.82	68.58	58.42
RefCOCO+	10,758	30.67	40.23	64.67	44.01
RefCOCOg	4,896	35.03	50.08	65.12	57.85

the paper; nothing was tuned on val. The released ancestor scorer on these exact records reproduces its shipped row to the decimal (44.82/40.23/50.08), which validates the swap end to end. Under this reading the transplanted rule scores 32.95/30.67/35.03 mIoU (Table 17), 8–15 points below the unguided released scorer and 30–36 below our box rule on the same signals. With pool, features, guidance and evaluation held fixed, the separation is a controlled measurement of the described rule as we read it — including its interaction with our feature geometry, since the published constants were fitted to theirs — and it is reported as such: it measures our reading of the published equations on our signals, not the original system, and no judgment of the original work is intended or supported. The released ancestor's same-records reproduction matches its shipped row to full precision (44.8248/38.16, 40.2289/33.5875, 50.0816/41.4453; recorded per dataset in the artifact), which validates the swap end to end.

6. Limitations

The same-class floors are cleared; the supervised cIoU level is the measured boundary. The training-free rule clears every same-class (zero-shot) published number on both metrics (50.14/74.66 vs. the best published zero-shot 44.40/43.30), and its gIoU sits just above the strongest supervised gIoU reference (74.34) by 0.32 (frozen rule: 0.26) — a parity-level margin, not a decisive exceedance — while the no-target decision is above the trained state of the art (N_{acc} = 85.70 vs. 79.70). The supervised cIoU level (68.94) is where the class stops; the corrected training-free cIoU (50.14) is 18.8 points below it. The earlier r-decomposition of this gap used the misaligned term map and is withdrawn rather than re-derived here. What survives is qualitative and measured independently of terms: the bind is partly perceptual (the forced-choice probe picks the GT-best of two candidate crops at 29.8% against 25% chance on the held-out grid, and at 59.3% on the two-object slice, below the 85.4% largest-candidate heuristic), so closing the gap needs a supervisor that selects boxes at near-oracle quality on two-object referents. Published cIoU numbers at that level come from models trained to select at that quality.

The gRefCOCO numbers carry a term-alignment correction. The gRefCOCO term-path evaluations in earlier drafts used a ref_id-keyed sentence map although the official evaluator scores each sentence separately; 84.7% of targeted rows in multi-sentence refs used another sentence's terms. After per-record alignment the frozen-parameter rule measures 49.63/74.60 (was 51.32/74.91); a held-out re-tune of its sxi scalars reaches 50.14/74.66 and is shipped. The declared-trained transfer recipe measures 49.40/74.26 (was 52.71/75.08) — below the rule. The recipe promotion, its bootstrap interval and the mechanism decomposition built on the same part tables are withdrawn; the shipped gRefCOCO row is the training-free rule. The RefCOCO-family results are unaffected (per-record text). The official metric arithmetic itself was re-executed from the released evaluator and matches ours, with one convention difference: an empty-prediction abstention contributes the GT area to the cIoU denominator, where the official targeted-false-positive case keeps a discarded mask's union.

The VLM is pretrained on COCO images. LLaVA-OneVision's pretraining data includes COCO (the source of the benchmark images), as is true for the MLLM that GLOD-RIS queries through its API; no referring annotations are used by either. This is the

standard training-free setting in this literature, but it is a weaker notion of "zero-shot" than a model that has never seen the image pool.

Which backbones are load-bearing: a measured weaker-lineup audit. The shipped lineup (SAM ViT-H, CLIP ViT-B/16, LLaVA-OneVision-7B boxes, the learned head; plus GroundingDINO on the gRefCOCO path only) was re-measured under strictly weaker replacements on the full val splits, each under a head refit on train-carved rows wherever the platform changed (backbone_audit.json). Two components can be weakened for free — one of them profitably: the box prior gets smaller and better (Qwen2.5-VL-3B, 3.8B, holds full-table SOTA and improves on the 7B boxes: 73.57/67.64/66.43 vs 70.57/66.35/65.51 mIoU — in this two-point comparison the advantage tracks the model family, not its size; a within-family size sweep is not measured), and replacing the released G2L region-text platform with our own plain masked-region CLIP features (no fusion, no complement blur) costs only 0.34/0.37/0.19 mIoU (70.22/65.98/65.32) while keeping SOTA. Two boundaries are real and measured: with SAM ViT-B (0.09B) the pool's own oracle drops from 79.1 to 71.4 mIoU on RefCOCOg and the refit head lands 57.57, 0.28 below the published GLOD-RIS floor — a margin that is within unquantified single-seed refit noise, so the load-bearing conclusion rests on the pool-oracle gap, not on the 0.28 (the ViT-L point between them is unmeasured: its original-format checkpoint was unreachable offline, its HF port restructure being incompatible); and the detector-free gRefCOCO realization (LLaVA boxes only, which necessarily drops the trained recipe: its candidate substrate is detector-derived) fails the same-class cIoU floor (38.56 vs 44.4) with the corrected per-record alignment. We additionally ran the rescue arms for that path — per-clause full-text boxes, presence- verified term filtering, the union/pair candidate family, a full-sentence box, and a 320-trial held-out search over the DINO-free configuration space — and none clears the floor (best held-out-searched clause configuration 40.56 cIoU / 70.56 gIoU; base noun 38.56, full-clause 37.95, presence-filtered 37.85, sentence box 36.70, pair/union 36.73): the detector-bearing path is load-bearing for cIoU on this benchmark, and the boundary is measured rather than assumed. The declared-trained no-detector head was not re-measured after the alignment correction; its earlier 28.29 is withdrawn with the other misaligned term-path numbers. The boundaries are reported per component, not tuned around. The pool-density axis was additionally varied (points_per_side $8 \to 16$, 110 % more proposals) and the diagnosis survives ($0.741 \to 0.720$).

Comparison protocol. The released HybridGL code and our reimplementation agree within 0.6 mIoU, and both sit 4.6 mIoU below the numbers printed in the HybridGL paper (§5.12). We therefore treat cross-paper absolute comparisons in this baseline family as cross-harness comparisons: every published row is quoted as published and labelled as such, and all our controlled comparisons are internal to one reproduction.

VLM latency. The box pass costs 0.6–1.8 s per expression on one RTX 4090, which is 3–10× the released pipeline's per-expression cost. The cost is cacheable (one box per unique expression) but not free, and the paper does not report a wall-clock comparison against the MLLM-based prior it replaces.

7. Conclusion

Training-free referring image segmentation has been optimised as a scoring problem, but at its operating point the pool is not the bottleneck — selection is, and the specific failure is localization. The released scores compare whole-region embeddings with whole-sentence embeddings; on descriptive expressions that signal is ambiguous across instances, and no stronger encoder repairs it. Adding one frozen, local, API-free component — a 7B vision-language model asked for the referent's bounding box, with the full expression as the prompt — converts selection into geometric matching and moves the three RefCOCO-family benchmarks substantially, under a protocol in which every fitted scalar comes from a held-out train-derived split and every reported number is a re-analysis of full-validation records.

We also report the boundary of the class we work in, together with a correction to our own gRefCOCO measurement. On gRefCOCO the no-target decision is solved above the trained state of the art ($N_{\text{acc}} = 85.70$ vs. 79.70), and the training-free rule beats every same-class (zero-shot) published number on both metrics (50.14/74.66 vs. 44.40/43.30), with gIoU just above the strongest supervised gIoU reference (74.34) at a parity-level margin. The supervised cIoU level (68.94) remains out of reach. gRefCOCO refs group several sentences and the official evaluator is per sentence; our earlier term paths looked terms up by ref_id, so most multi-sentence rows used another sentence's terms. After per-record alignment, with frozen parameters, the training-free rule measures 49.63/74.60; a held-out re-tune of its sxi scalars reaches 50.14/74.66 and is shipped. The declared-trained transfer recipe measures

49.40/74.26. The recipe promotion of an earlier draft, its bootstrap and its mechanism account are withdrawn as artifacts of the misalignment; the shipped row is the training-free rule. Finally, a comparison-protocol note: the released HybridGL code and our reimplementation agree within 0.6 mIoU, and both sit 4.6 mIoU below the numbers printed in the HybridGL paper; for this baseline family we therefore treat cross-paper absolute comparisons as cross-harness comparisons — every published row is quoted as published and every comparison against it is labelled as such — and keep our controlled comparisons internal to one reproduction.

A. Appendix: implementation details

Scoring pass. For every expression we store: the released semantic score under four text variants (full expression, spatial-stripped, content-only, head noun phrase), the negative-noun score, the released GEM coherence score, per-proposal mask area, centroid and bounding box, the strict-containment parent of every proposal, the pool pairwise IoU matrix, and the per-proposal IoU against ground truth (the union of the reference's masks on gRefCOCO). Everything else in this paper is a re-analysis of these records; the records themselves come from one forward pass per dataset (exp/exp4_guided_full.py, exp/exp21_grefcoco_full.py).

Shipped configuration. Candidate base = $\mathrm{rank}(S^{\mathrm{dec}}) + \gamma\,\mathrm{rank}(S^{\mathrm{coh}})$ with $\gamma = 0.49$; $k = 15$; overlap mode = IoU; selection = argmax of the mode over the candidates; growth off. These five values are the best of 300 Optuna trials on the held-out split, objective = minimum over datasets of mIoU (exp/out_final/optuna_study_sent.json); the noun-phrase study is the second file. The earlier repairs (relation-gated operator with $k = 8$ and a two-axis gate, containment growth at $|A \cap B|/|A| \geq 0.95$) are reported as ablations and are off in the shipped configuration.

Bounding-box prior. Prompt: "Locate the {expression} in this image. Reply with ONLY the bounding box as [x0,y0,x1,y1] in pixel coordinates measured from the top-left corner, or the single word none if it does not appear in the image.", with the full expression substituted. Decoding is greedy, max_new_tokens= 32; the reply is parsed for its first four numbers and values ≤ 1.5 are read as normalised coordinates. Boxes are cached per (image, expression) under exp/out_vlm_boxes/; the vision tower pass is cached per image. The 8-bit backend (exp/convert_llavaov_hf.py converts the cached checkpoint to the HF layout so the standard bitsandbytes path applies) produced the same boxes as bf16 on our agreement sample.

GRES extension. Clause split on top-level "and"/"or"/","; one box per clause; per-clause selection by Eq. 1; union of the selected masks; empty output when every clause answers none.

Compute. All scoring passes ran on 4×RTX 4090. Per dataset the pass costs about 0.9–1.2 GPU-hours; the entire paper (four datasets, plus controls and ablations, which are CPU re-analyses) is under 6 GPU-hours.

Val-pass ledger and reproduction pins. Every arm in the learned-prior table and the transfer-inversion study is a single frozen application of the full official val split. Within the transfer-inversion study that is five new configurations (recipe, recipe+combination policy, strata-$\{1, 2\}$, weights-only, tau-only), seven pre-registered leave-one-rare-ref-out refits (diagnostics only; the shipped row was declared unchanged on any outcome), and two bit-reproduction passes for the cluster bootstrap; the weak-signal democracy probe ran on a seeded 800-row val sample under its own single-pick protocol (its number is a sample mIoU, never a full-val cIoU). Arm, threshold and weight selection used the held-out train carve-out only. Reproducing the shipped recipe bit-exactly requires OMP_NUM_THREADS=4 (as the shipped runs used): under a different thread count the logistic-regression fits drift in the last decimal, two margin-boundary rows flip their rank-1 second part, and the train table rebuilds with 10,352 parts instead of 10,351 (val 52.684 vs 52.706); under the pin an independent control refit reproduces the shipped row exactly.

B. Appendix: per-dataset detail

Taxonomy per dataset. Table 1 reports the oracle-gap taxonomy on all four benchmarks; the wrong-instance share is stable across datasets and across two scorers that differ by 10 mIoU.

Strata and quartiles. On RefCOCO val the operator's gain decomposes by relation type: horizontal $n = 4{,}749$ +0.1169, vertical $n = 443$ +0.0735, middle $n = 584$ +0.0820, ordinal (abstains) and no-relation (abstains) zero by construction. By ground-truth area quartile the operator is positive in all four (+0.0427 / +0.0553 / +0.0566 / +0.0801), while the growth control's gain is confined to the largest.

gRefCOCO rows. Table 7 lists every row of the gRefCOCO evaluation, including the margin sweep for plural expansion and the two controls (no relation gate, no growth).

Term-alignment audit and the official metric. gRefCOCO refs group several sentences (14,229 val records over 4,237 ref_ids) and the official evaluator scores each sentence separately. We imported the released ReLA ReferEvaluator with a minimal detectron2 stub and ran a five-case battery (targeted exact/wrong/abstain-FP, no-target TP/FN); its accumulated cIoU/gIoU/N-acc/T-acc arithmetic matches ours, with one convention difference: a targeted false positive in the official path keeps the discarded mask's union in the cIoU denominator, while our empty-prediction abstention contributes only the GT area. The audit also showed that our gRefCOCO term lookups were keyed by ref_id rather than by the row's own sentence, so 84.7% of targeted rows in multi-sentence refs used another sentence's terms. All gRefCOCO term-path results in this paper are re-evaluated with per-record alignment (gref_tf_aligned.json, gref_transfer_recipe_aligned.json, gref_search_val_part0.json); the RefCOCO-family path uses per-record text and is unaffected. The corrected frozen-parameter rule is 49.63/74.60 cIoU/gIoU; a held-out re-tune of the sxi scalars reaches 50.14/74.66 and is shipped. The declared-trained recipe measures 49.40/74.26, so it is not shipped and the earlier promotion is withdrawn.

Published-number transcription. Published rows in Table 5 were transcribed from the GLOD-RIS Table I and HybridGL Table 1 PDFs (both in the repository under papers/) and cross-checked between the two sources where they overlap.

C. Appendix: qualitative examples

D. Appendix: evidence map

Every claim in the paper maps to a stored artifact: exp/out_final/main_{refcoco,refcocoplus,refcocog,grefcoco}.json (one per dataset; all rows, CIs, strata, quartiles, taxonomy); exp/out_final/gref_topk_reselection.json and gref_learned_selector_transfer.json (change 16 and the documented negative transfer result); exp/out_final/protocol_audit.json (the metric-protocol audit of §5.1: per-dataset protocol source, metric definition, and the re-evaluation of the saved predictions under it, with the numbers after alignment — including the completed gRefCOCO coverage and the pooled oIoU); exp/out_final/ablation.json (each row with its own evidence file); exp/out_final/sota_table.json (measured rows carry evidence paths; published rows carry transcription sources); refine-logs/METHOD_GRAVEYARD.md (killed methods and the control that killed each); paper/figures/specs/architecture.json and the rendered architecture.pdf (deterministic vector figure); exp/out_extreme/records_*.jsonl (the scoring passes every analysis reads); exp/out_final/headroom.json (the structural cause: top-3 containment, oracle rank, category-score top-8); exp/out_final/cross_dataset.json (the dose-response and the residual strata); exp/out_final/pool_robustness.json (points-per-side 8 vs 16); exp/out_final/evidence/repro_blocks.json with the raw logs repro_official_blockA.log and repro_official_blockB.log (the two disjoint reproduction-audit blocks); exp/out_final/optuna_study_{noun,sent}.json (the two 300-trial studies and their per-dataset held-out mIoU); exp/tuning/*_refs.json (the held-out split: seeded train refs); exp/out_vlm_boxes/ (one cached VLM box per expression); exp/out_final/gref_ceiling.json (the gRefCOCO no-target wall: oracle set-IoU, the leaky logistic bound, and the N-acc requirement at perfect masks); exp/out_final/exp37_{refcoco,refcocog}_gt.json and exp37_vlm_sample800.json (the box-prior ceiling and its realistic-box sample); exp/out_final/optuna_study_gref{2,3,4,5}.json (the gRefCOCO held-out studies: hand rules, learned selector, decision rules, clause features; 300 trials each); exp/out_final/gref_exists_probe_sh*.json and the exists cache namespace (the single yes/no presence probe); exp/out_final/gref_ensemble_{roc,val}.json, gref_rule_showdown.json, and gref_ensemble_aggregates.json (the three-prompt presence ensemble: held-out rule selection, full-val frontier, and the exact union-mass requirements behind the shipped row); exp/out_final/gref_wall.json (the three-axis wall analysis and the exact oracle-realization requirement); exp/out_final/gref_tuned_val2000.json and exp/out_final/gref_full_val.json (the frozen gRefCOCO model's validation evaluation); exp/out_final/component_provenance.json with head_rebuild.json, head_rebuild_parts/*.json and exp159b_boxsource_probe.json (the component-provenance audit of §5.4: per-component provenance, the head-rebuild arms, and the box-source/phrase probes used to reconcile the quoted attribution); exp/out_final/test_columns_learned.json with exp/out_test_columns/learned_*.json (the

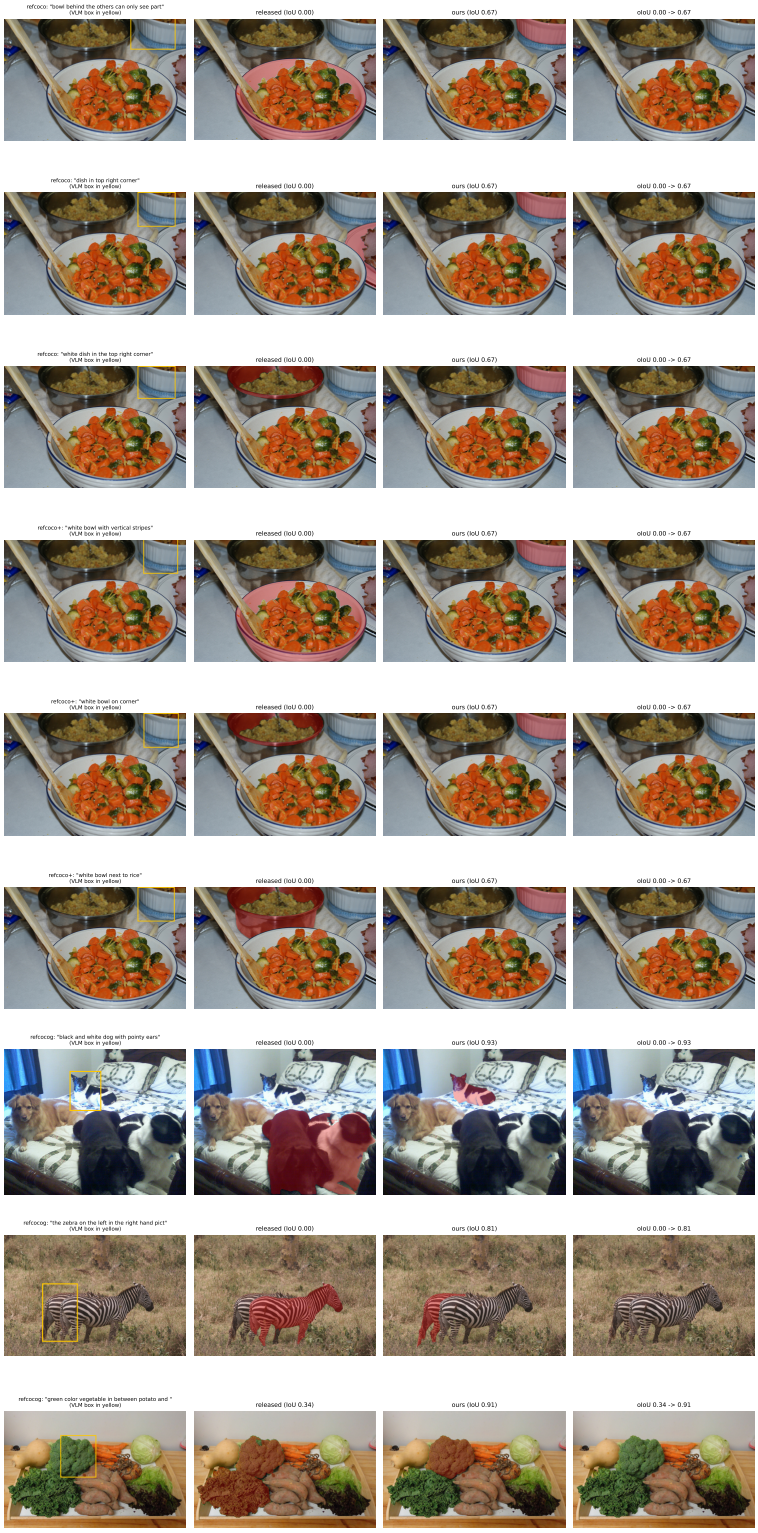


Figure 3. Examples where the box prior changes the selection (one per dataset family, selected by a criterion fixed in exp/exp45_qualitative.py: the method and the released pipeline disagree and the IoU gap exceeds 0.25). Column 1 shows the VLM box (yellow) overlaid on the image and the expression; columns 2–3 the released selection and ours. Ground-truth IoU is in each title. These are illustrative, not a random sample.

promoted head's test-column re-evaluation); exp/out_final/backbone_audit.json with backbone_swaps/*.json and gref_no_detector_val.json (the weaker-lineup audit of §6: per-component swap arms on the full val splits, each under a head refit on train-carved rows where the platform changed).

Component provenance (component_provenance.json). Each shipped component is tagged by origin. Ours: the VLM box prior and box-fill pick; the gRefCOCO multi-term machinery, the three-prompt existence ensemble, and the declared-trained acceptance prior; the learned selector head; the protocol alignment. Released HybridGL: the masked-region feature extractor (modified CLIP ViT-B/16 with the G2L fusion), the released semantic score (0.5 sentence + 0.5 noun-phrase ensemble), the GEM coherence score, the fused selection, and the released-score candidate restriction — used by the training-free reference realization and, as a feature path only, by the promoted head's region-text term. Generic frozen: the SAM ViT-H pool and the GroundingDINO detector gate. The head rebuild behind the promotion, and both drop-one deltas (restriction-given-box $-1.45/-0.56/+0.29$; box-given-restriction $+25.2/+26.8/+17.6$ mIoU), are reported in Table 9; an independently quoted box-over-full-pool figure (50.20/40.24/49.48) could not be reproduced under any tested configuration (overlap mode, box phrase, box source) and is recorded as unreconciled in the artifact; the measured drop-one deltas above stand.

References

[1] Nico Catalano et al. Mars: a multimodal alignment and ranking system for few-shot segmentation. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026.

[2] Sahar Kazemzadeh, Vicente Ordonez, Mark Matten, and Tamara L. Berg. Referitgame: Referring to objects in photographs of natural scenes. In Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), 2014.

[3] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023.

[4] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. Llava-onevision: Easy visual task transfer. arXiv preprint arXiv:2408.03326, 2024.

[5] Jiachen Li, Qing Xie, Renshu Gu, Jinyu Xu, et al. Lgd: Leveraging generative descriptions for zero-shot referring image segmentation. Pattern Recognition, 2025.

[6] Wenhui Li, Chao Pang, et al. Bidirectional mask selection for zero-shot referring image segmentation. IEEE Transactions on Circuits and Systems for Video Technology, 35(1), 2025.

[7] Chang Liu, Henghui Ding, Xudong Jiang, et al. Gres: Generalized referring expression segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

[8] Ting Liu and Siyuan Li. Hybrid global-local representation with augmented spatial guidance for zero-shot referring image segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 29634–29643, 2025.

[9] Junhua Mao, Jonathan Huang, Alexander Toshev, Oana Camburu, Alan L. Yuille, and Kevin Murphy. Generation and comprehension of unambiguous object descriptions. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2016.

[10] Minheng Ni, Yabo Zhang, Ke Feng, Xiujun Li, Daxin Guo, and Wangmeng Zuo. Ref-diff: Zero-shot referring image segmentation with generative models. In arXiv preprint arXiv:2308.16777, 2023.

[11] Yucheng Suo, Linchao Zhu, and Yi Yang. Text augmented spatial-aware zero-shot referring image segmentation. In Findings of the Association for Computational Linguistics: EMNLP 2023, 2023.

[12] Jingchao Wang, Zhijian Wu, Dingjiang Huang, Yefeng Zheng, and Hong Wang. Unlocking the potential of mllms in referring expression segmentation via a light-weight mask decoder. In arXiv preprint arXiv:2508.04107, 2025.

[13] Junjie Wu et al. Glod-ris: Zero-shot referring image segmentation with gaussian localization and orthogonal decoupling. IEEE Transactions on Circuits and Systems for Video Technology, 2026.

[14] Xiaoyang Xu et al. E-sam: Training-free segment every entity model. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025.

[15] Seonghoon Yu, Paul Hongsuck Seo, and Jeany Son. Zero-shot referring image segmentation with global-local context features. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

[16] Yiwu Zhong, Jianwei Yang, Peng Wang, Ronghui Tang, Cheng-Lin Zhao, Renjie Liu, Xiang Li, and Yuhui Yuan. Proposalclip: Unsupervised open-category object proposal generation via exploiting clip cues. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

[17] Chong Zhou, Chen Change Loy, and Bo Dai. Extract free dense labels from CLIP. In Proceedings of the European Conference on Computer Vision (ECCV), 2022.